

# Tunisia, audited — Cut the Dose

The burden measured, the cheapest cut in each chain, the 90-day pilot, the public scoreboard, and the food stack — plain words, full arithmetic.

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## 1. The burden, measured — and what the numbers say

Roughly **379 years of healthy life lost every year** in the basin (30-70k people): 200 to fluoride — teeth and bones — 120 to heavy metals, 59 to dust (H4-H6 anchors, DERIVED). A child in Redeyef develops asthma at age two; the nearest hospital is 70 km; the doctors' prescription is *leave Gafsa* (H16).

## 2. Cut the Dose — the cheapest link in each chain

| CHAIN                           | THE CUT                                                              | COST                                                                                |
|---------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Fluoride: water → teeth → bones | point-of-use filters (Nalgonda/bone char)                            | <b>\$2,592 per healthy year</b> vs the \$4,000 standard — this rung pays for itself |
| Metals: water → gut → kidney    | zinc, selenium, probiotics (gut binding)                             | in the \$2.20M/yr full stack                                                        |
| Dust: air → lungs               | clean-air boxes + N95-class masks (98%+ vs ≤44% cloth)               | RCT: poorly-controlled asthma IRR 0.43-0.45                                         |
| Fluorosis in children           | calcium + vitamin C + D3 — the combination shown to REVERSE it (RCT) | pennies a day                                                                       |

## 3. The 90-day pilot — and the scoreboard that follows

**\$58,000:** 100 households + 2 schools, 90 days, results published. **Gate:** filter uptime ≥70%, urinary F ↓20%. **The scoreboard:** water chemistry, filter uptime, cohort KPIs — public, every quarter. The health data of the basin becomes the scoreboard of the nation — and the collateral that lowers the price of every loan in the plan (100-150bp on ~\$800M of portfolio debt: \$8-12M/yr).

## 4. The food stack — protein the people can pay

| PIECE           | NUMBER                                                                                       |
|-----------------|----------------------------------------------------------------------------------------------|
| Spirulina ponds | 10 ha → 130 t/yr → <b>178,082 children</b> at the RCT dose (2 g/day); the basin needs 0.7 ha |
| Protein price   | \$5-10/kg where imported meat protein costs \$15-25/kg                                       |

| PIECE                 | NUMBER                                                                                                                        |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Terfas domestic quota | ≥30% of harvest sold at home at ≤€12/kg — written into the contracts                                                          |
| Bone char             | 90 t/yr from local slaughterhouses → the filters, and a workshop of jobs                                                      |
| Success metric        | the price of protein and vegetables in Gafsa and Gabès bends downward within 24 months of first harvest — published quarterly |

## 5. The two futures, in plain words

**Without the plan:** the dose accumulates — fluoride in the teeth before age 8, cadmium in the kidneys for decades, dust in the lungs. The fittest leave: engineer to Paris, founder to London. Every departure makes the basin harder for those who stay. That future requires no decision — it is the default.

**With the plan:** we change the environment, not the genes. Clean water → intact enamel and strong bones within one generation. Protein → nourished brains. Jobs → a reason to stay. Same people, same geography — a different fitness landscape. The plan is evolution by other means.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

## Works cited

- [H1] Guissouma 2017 (Chemosphere): Tunisia tap-water fluoride 0-2.4 mg/L; 25% of population at dental-fluorosis risk, 20% at skeletal-fluorosis risk (WHO limits) — [https://pubmed.ncbi.nlm.nih.gov/28284958/...](https://pubmed.ncbi.nlm.nih.gov/28284958/)
- [H2] Bouchiba 2026 (Toxics 14(5):438): Mdhilla beneficiation effluent Cd 0.49, Pb 0.71, Cr 3.09 mg/L — exceeding discharge limits; risk driven by Cd, Co, Tl — <https://doi.org/10.3390/toxics14050438...>
- [H3] Gupta 1996 RCT: fluorosis REVERSED in children (Ca + VitD3 + ascorbic acid; n=25; water 4.5 ppm F; double-blind) — [https://pubmed.ncbi.nlm.nih.gov/8942013/...](https://pubmed.ncbi.nlm.nih.gov/8942013/)
- [H4] Ghaemi 2024 (Iran): dental-fluorosis DALY rate up to 981.45 (UI 353.23-1618.40) per 100,000 in a high-F district; Monte Carlo dose-response — [https://pubmed.ncbi.nlm.nih.gov/39003868/...](https://pubmed.ncbi.nlm.nih.gov/39003868/)
- [H5] Abtahi 2019 (Water Research): age-sex specific DALYs from drinking-water fluoride, Iran national study — [https://pubmed.ncbi.nlm.nih.gov/30953859/...](https://pubmed.ncbi.nlm.nih.gov/30953859/)
- [H6] Naddafi 2022 (Env Research): burden of disease from heavy metals in drinking water, Iran 2019 — [https://pubmed.ncbi.nlm.nih.gov/34529973/...](https://pubmed.ncbi.nlm.nih.gov/34529973/)
- [H7] Egner 2014 RCT (n=291): broccoli-sprout beverage 12 wks -> urinary benzene mercapturic acid +61%, acrolein +23% (P≤0.01); GSTT1-positive excrete more — [https://pubmed.ncbi.nlm.nih.gov/24913818/...](https://pubmed.ncbi.nlm.nih.gov/24913818/)
- [H8] Pabis 2018: zinc supplementation -> >30% reduction of kidney/liver Cd in aged metallothionein-transgenic mice — [https://pubmed.ncbi.nlm.nih.gov/30103140/...](https://pubmed.ncbi.nlm.nih.gov/30103140/)
- [H9] Zhai 2016: Lactobacillus plantarum CCFM8610 binds gut Cd, protects the intestinal barrier, lowers tissue Cd in mice — [https://pubmed.ncbi.nlm.nih.gov/27208136/...](https://pubmed.ncbi.nlm.nih.gov/27208136/)
- [H10] Mueller 2018: N95-class respirators ≥98% filtration efficiency vs ≤44% for any cloth — [https://pubmed.ncbi.nlm.nih.gov/29779694/...](https://pubmed.ncbi.nlm.nih.gov/29779694/)
- [H11] Allen & Barn 2020: HEPA purifiers substantially cut indoor PM2.5 and improve subclinical cardiopulmonary health; population benefit often exceed costs — [https://pubmed.ncbi.nlm.nih.gov/33241434/...](https://pubmed.ncbi.nlm.nih.gov/33241434/)
- [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — [https://pubmed.ncbi.nlm.nih.gov/34980119/...](https://pubmed.ncbi.nlm.nih.gov/34980119/)
- [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — [https://pubmed.ncbi.nlm.nih.gov/36476193/...](https://pubmed.ncbi.nlm.nih.gov/36476193/)

14. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — <https://ejatlas.org/conflict/phosphate-mining-in-gafsa...>
15. [H15] NRGi: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — <https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...>
16. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — <https://geographical.co.uk/science-environment/gafsa-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...>
17. [Rc7] Terfezia cultivation: commercial since 1999-2001 (Murcia, Spain); fruiting at year 2-3; 20-yr studies: average ~376 kg/ha, peak 1,069 kg/ha, drought years 0-2 kg/ha — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8907101/...>
18. [Rc9] Terfezia farm-gate €20-60/kg (€200/kg only in extreme scarcity in Gulf markets); Tuber melanosporum €700-1,500/kg retail — different commodities — <https://trufamania.com/desert-truffles.htm...>
19. [Rc18] Desert-truffle market depth: Mamora (Morocco) wild harvest peaks ~100 t/yr; luxury band breaks past ~1,000-1,500 t/yr of new supply — <https://www.mdpi.com/1999-4907/14/5/952...>